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How does the market respond to textual novelty: Joint analysis of similarity and sentiment on Japanese corporate disclosures

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ABSTRACT

Market reactions to corporate disclosures have been widely studied, yet not all statements within them are equally informative. We investigate the role of informational novelty within disclosures and its impact on the market, focusing on Japanese financial results briefings. We measure sentence-level sentiment and novelty using a semantic approach. Our results show that sentiment in novel (low-similarity) statements has stronger explanatory power for post-event stock returns than sentiment in sticky (high-similarity) statements. A framework that jointly analyzes similarity and sentiment highlights informative differences within disclosures often overlooked by document-level approaches, offering a broadly applicable basis for financial text analysis.

1. Introduction

Firms communicate with markets through multiple channels such as financial statements, annual reports, and earnings conferences. These disclosures help investors assess firm's performance and prospects, and their market impact has been widely studied.

Among these disclosures, not all disclosure content is equally informative. Reused or boilerplate statements are often associated with lower informational value, and are linked to reduced usefulness for investors in Form 10-K filings (Brown and Tucker, 2011). By contrast, Cohen et al. (2020) show that textual changes in 10-K filings explain subsequent stock returns: larger changes are followed by lower returns. These findings suggest that informational values vary with novelty, that is, whether the market has already processed the content. Much of the literature, however, aggregates sentiment at the document, firm, or daily level, without distinguishing the novelty of individual statements (Tetlock, 2007; Henry, 2008; Kothari et al., 2009; Feldman et al., 2010; Davis and Tama-Sweet, 2012; Keith and Stent, 2019; Deveikyte et al., 2022; Fatouros et al., 2023).

This study investigates the market value of novel information by approximating novelty with semantic similarity to prior disclosures. We analyze Japanese financial results briefings: firm-specific, non-mandatory, and oral presentations after earnings announcements. In mandatory filings, disclosure content is often prescribed by regulatory requirements, which can lead to increases in boilerplate, stickiness, and redundancy as discussed on 10-K filings (Dyer et al., 2017). The briefings are not subject to prescribed requirements and are delivered orally, so they are expected to involve less word-for-word reuse of previous presentations. These features make briefings more suitable for measuring informational novelty.

We employ modern natural-language processing to address the colloquial briefing transcripts. For sentiment, we fine-tune a Japanese financial BERT model (Suzuki et al., 2023); for similarity, we use a model based on Sentence-BERT (Reimers and Gurevych, 2019) to obtain sentence embeddings. These methods provide a robust approach to paraphrasing and stylistic variation.

Our main finding is that novel statements have distinct explanatory power for post-event returns. Sentiment in dissimilar sentences explains post-event returns significantly better than sentiment in similar sentences. We confirm that sentence-level novelty filters information that is useful for predicting market responses, which is a feature missed by document-level approaches. In addition, the method highlights the most informative parts of disclosures, offering a potential countermeasure to concerns about declining usefulness of disclosure documents (e.g., the MD&A in 10-K filings discussed by Brown and Tucker, 2011).

2. Methodology

2.1. Data

We analyze presentation transcripts from financial results briefings of Japanese firms, sourced from logmi Finance[1] until October 18,

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[1] https://finance.logmi.jp/

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2024. We collect related metadata, and manually correct them when needed. The financial results briefings (hereafter, "events") are filtered as follows. We exclude financials and banks (TOPIX 17 sectors) because their financial statements are not directly comparable to other sectors. We link an event to a financial statement when the event date is within 15 days after the earnings announcement. Our analysis requires fundamental factors for both the reported period and the same period in the prior year. We require stock return data for trading days $[-130, 10]$ relative to the event date. Because we approximate textual novelty relative to prior events, we drop each firm's first event. These filters yield 3045 events from 703 firms, covering both large- and small-cap firms: 1634 events with market value of equity above 100 billion yen and 1411 events below that threshold.

We normalize fundamental factors to control for firm scale. For each event i, we define sales growth and margin change as:

$$\Delta\text{Sales}_i = \frac{\text{Sales}_i}{\text{Sales}_i^*} - 1, \tag{1}$$

$$\Delta\text{Margin}_i = \frac{\text{OperatingIncome}_i}{\text{Sales}_i} - \frac{\text{OperatingIncome}_i^*}{\text{Sales}_i^*}, \tag{2}$$

where Sales_i and OperatingIncome_i refer to the period discussed at event i, and the asterisk denotes the corresponding values in the prior year. We winsorize outliers of ΔSales_i and ΔMargin_i at the 1st and 99th percentiles.

2.2. Related literature and hypothesis

Textual modifications in firm disclosures convey information because they reflect managers' choices about what to communicate. Cohen et al. (2020) show that larger textual changes in 10-K filings are associated with lower subsequent stock returns. They claim that "86% of changes consist of 'negative' sentiment changes." in Form 10-K possibly due to litigation risk from failing to report negative information. In contrast, unchanged expressions are often discussed in relation to the reuse of past descriptions (stickiness) or redundant expressions (boilerplate) (Dyer et al., 2017). In 10-K filings, boilerplate is associated with reduced informational usefulness (Brown and Tucker, 2011). Thus, modified and sticky texts likely convey different meanings within the same disclosure.

However, sentiment analysis often treats both types of statements equally. For example, Loughran and McDonald (2011) measure tone in 10-K filings using a finance-specific sentiment dictionary and show that negative word usage is significantly related to returns. Other studies also compute overall sentiment without distinguishing the modification of individual statements (Tetlock, 2007; Henry, 2008; Kothari et al., 2009; Feldman et al., 2010; Davis and Tama-Sweet, 2012; Keith and Stent, 2019; Deveikyte et al., 2022; Fatouros et al., 2023). Even though novel information can change future prospects of the investors, the role of novelty remains underexplored in sentiment analysis. Since it is difficult to assess novelty against all information sources, we approximate it by measuring sentence-level similarity to prior disclosures. We apply this approach to sentiment analysis and test how novel and sticky sentences relate to market reactions. Based on these ideas, we test:

Hypothesis. The sentiment of novel (dissimilar) sentences has a greater impact on stock returns than the sentiment of sticky (similar) sentences.

We test the hypothesis using financial results briefings by Japanese firms. Conference calls of U.S. firms have been widely studied; for example, Matsumoto et al. (2011) analyze abnormal absolute returns around announcements, and Brockman et al. (2015) examine the impact of tone on returns. The informational content of Japanese financial results briefings has also been studied (Manabe et al., 2022; Kuroki et al., 2023; Nakatsuka and Suimon, 2024). Although these literature confirmed the importance of conference calls or financial results briefings, the role of novelty has not been examined.

2.3. Similarity and sentiment

We approximate novelty as sentence-level semantic similarity to the prior event. Previous studies often estimate similarity by comparing sequences of words (Dyer et al., 2017) or word frequencies (Cohen et al., 2020). To account for lexical variation, we employ a semantic approach based on Japanese Sentence-BERT, the GLuCoSE model.[2] For sentence d, we evaluate the cosine similarity between embedded unit vectors $V(d)$ as

$$\text{Sim}(d) = \max(\{V(d) \cdot V(d_j)\}_{d_j \in D'}), \tag{3}$$

where D' is a set of sentences split by periods from the prior event transcript. High similarity sentences include recurring statements such as financial items, segment names, and firm's slogans. Conversely, low similarity sentences include non-recurring statements such as firm-specific actions, services, products, people, or place. Thus, similarity helps separate boilerplate from novel content. To mitigate firm- or speaker-style effects, we split each transcript into 'similar' (D_{similar}) and 'dissimilar' ($D_{\text{dissimilar}}$) sentences at its median similarity.

We then conduct sentence-level sentiment analysis. Sentiment analysis typically uses dictionaries (Turney, 2002) or machine learning methods (Pang et al., 2002) including modern BERT models (Devlin et al., 2019) and FinBERT in the financial domain Araci (2019), Liu et al. (2020), Yang et al. (2020). We start from a BERT model pretrained on Japanese financial texts (Suzuki et al., 2023),[3] and fine-tune it on the chABSA dataset provided by TIS Inc.[4] The dataset contains sentences from Japanese securities reports in 2016 labeled as positive or negative for business performance aspects (e.g., sales, profits). The sentiment for event i is then evaluated as:

$$S(D_i) = \frac{\overline{p}_{\text{pos}}(D_i) - \overline{p}_{\text{neg}}(D_i)}{\overline{p}_{\text{pos}}(D_i) + \overline{p}_{\text{neg}}(D_i)}, \tag{4}$$

where $\overline{p}_{\text{pol}}(D_i)$ is the average predicted probability over sentences in D_i for pol = pos or neg.

Similarity is expected to provide incremental information for sentiment analysis. To assess this, we examine its relation with sentiment and find only a weak correlation: at the sentence level, its correlation with the predicted probability of positive sentiment is 0.12, and with the predicted probability of negative sentiment is 0.09. Moreover, the distributions of sentiment are shown in Fig. 1, where sentences are divided into similar and dissimilar groups based on the median similarity across all sentences. The horizontal axis represents the difference between the predicted probabilities of positive and negative sentiment. Similar sentences have more polarized sentiment scores than dissimilar sentences. Accordingly, we incorporate similarity in sentiment analysis, as it may capture information independent of sentiment. We define sentiment for similar and dissimilar subsets as $S(D_{i,\text{similar}})$ and $S(D_{i,\text{dissimilar}})$, and the standardized sentiment differential as

$$\Delta S_i = \frac{S(D_{i,\text{dissimilar}})}{\sigma\left[S(D_{i,\text{dissimilar}})\right]} - \frac{S(D_{i,\text{similar}})}{\sigma\left[S(D_{i,\text{similar}})\right]}, \tag{5}$$

where each sentiment is normalized by its standard deviation across events.

2.4. Method

We conduct an event study to test the market impact of novel information. Abnormal returns (ARs) is introduced to remove overall market trends and stock-specific characteristics. We construct the Fama–French three-factor model for the Japanese stock market following Kubota and Takehara (2007). Exposures to market, size and value factors are estimated over a 120-trading-day window $[-130, -11]$. Cumulated ARs (CARs) are defined as $\text{CAR}_{i,t_1,t_2} = \sum_{t=t_1}^{t_2} \text{AR}_{i,t}$.

[2] https://huggingface.co/pkshatech/GLuCoSE-base-ja-v2
[3] https://huggingface.co/izumi-lab/bert-base-japanese-fin-additional
[4] https://github.com/chakki-works/chABSA-dataset

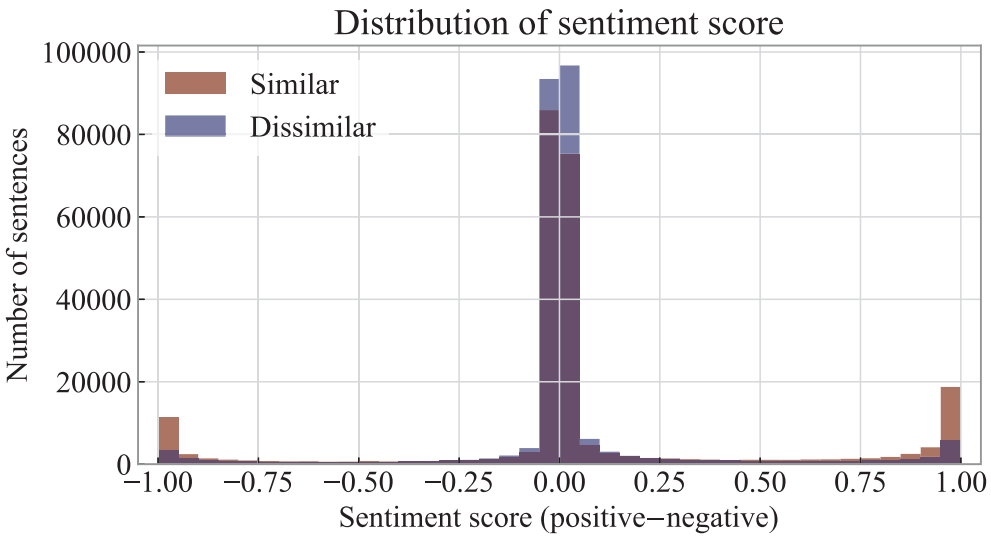


Fig. 1. Sentiment scores for the similar and dissimilar sentence groups.

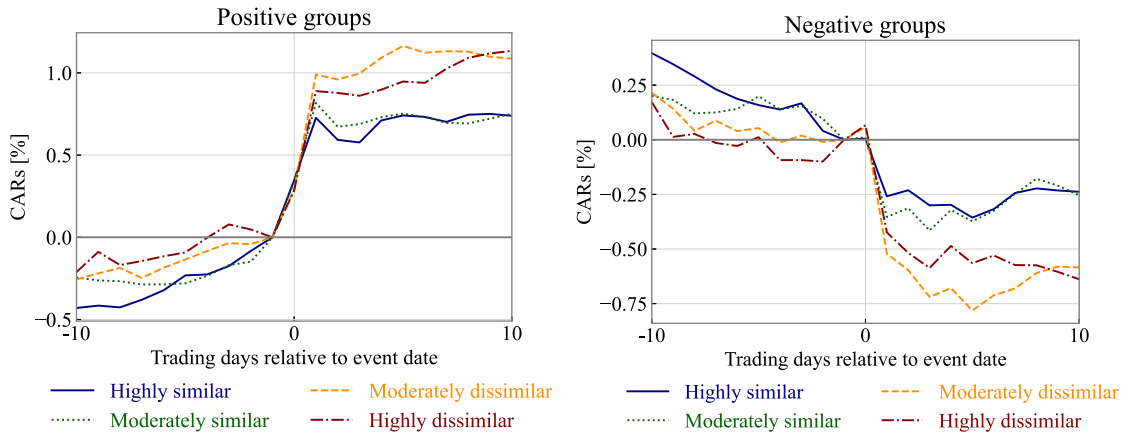


Fig. 2. Mean CARs around events by sentiment for sentences with different similarity.
Note: Financial results briefings are split into positive and negative sentiment groups, with sentiment on sentences separated based on similarity, from highly similar sentences to highly dissimilar sentences. The left (right) panel shows mean CARs for positive (negative) events.

We test the relation between sentiment and CARs using Welch's t-test. Events are split into two groups: positive events U with the top 50% by sentiment, and negative events D with the bottom 50%.

Next, we conduct regression analysis on five-day post-event CARs to test hypothesis:

$$\mathrm{CAR}_{i,0,4} = \alpha + \beta_{\mathrm{dissimilar}} S(D_{i,\mathrm{dissimilar}}) + \beta_{\mathrm{similar}} S(D_{i,\mathrm{similar}}) + \beta_{\mathrm{sim}} \mathrm{Sim}_i + \sum_X \beta_X X_i + \sum_s \gamma_s B_{s,i} + \sum_y \gamma'_y B_{y,i} + \epsilon_i, \quad (6)$$

where β denotes regression coefficients. Sim_i is the average similarity across sentences for event i. Controls X include ΔMargin, ΔSales, LogMKT for the log of the market value, PBR for a price to book ratio, respectively. We include industry dummies $B_{s,i}$ based on TOPIX 17 sectors and period dummies $B_{y,i}$ (year-quarter).

3. Results and discussion

ARs are compared across sentiment-sorted event groups. To examine how event groups vary with similarity, sentences are partitioned into four quartile-based groups-highly similar, moderately similar, moderately dissimilar, and highly dissimilar. For each group, we construct an event-level sentiment measure using only sentences in that similarity group, and then sort events into positive and negative groups based on this measure. Mean CARs around the event date are shown in Fig. 2, which indicates that sentiment from dissimilar sentences tends to explain post-event CARs more effectively than sentiment from similar sentences. These results are also observed for both large and small firms in Fig. 3, where sentiment is evaluated separately for sticky (similar) and novel (dissimilar) sentences within transcripts.

Pre-event returns also reflect sentiment derived from sticky sentences, possibly because such information is partially contained in, or can be inferred from, other information available before the event, for example through prior disclosures such as financial result announcements. Consistent with this interpretation, Table 1 shows that sentiment in similar sentences is more strongly correlated with the fundamental factors in Eqs. (1) and (2) than sentiment in dissimilar sentences. This effect is pronounced for small-cap firms, consistent with Campbell et al. (2021), which investigated pre-event abnormal trading volume around Form 8-K filings.

These differences are tested by using Welch's t-test shown in Table 2. These groups exhibit statistically significant differences in ARs within a few days after the event. Pre-event CARs in the $[-5,-1]$ window are significant for similar sentences but not for dissimilar sentences. This suggests that similar sentences likely consist of known statements already reflected in stock prices. By contrast, dissimilar sentences provide information insufficiently reflected beforehand and thus have larger

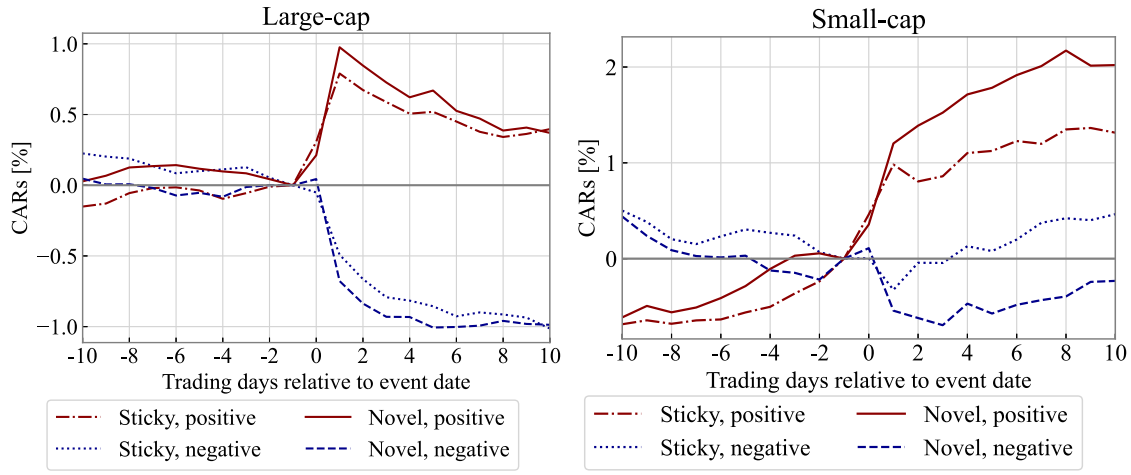


Fig. 3. Mean CARs around events by sentiment, sentence similarity, and firm size.
Note: Financial results briefings are split into positive and negative sentiment groups, with sentiment evaluated separately for sticky (similar) and novel (dissimilar) parts of transcripts. The left (right) panel shows CARs for large-cap (small-cap) firms.

Table 1
Correlation for sentiment and fundamental factors.

Sentiment	ΔSales_i	ΔMargin_i
Similar	0.39	0.24
Dissimilar	0.24	0.17

Note: The table shows the correlation coefficient between sentiment and fundamental factors across all events.

Table 2
Statistical t-test for differences in ARs and CARs.

AR	Sentiment		
(CAR)	Similar sentences	Dissimilar sentences	ΔS(Dissimilar−Similar)
0	2.06**	1.48	0.36
1	4.24***	6.44***	2.50**
2	−2.03**	0.65	2.21**
3	1.22	1.34	−0.19
4	0.45	0.00	−1.01
5	0.20	1.77*	1.64
[−5, −1]	2.63***	1.04	−0.14
[0, 4]	3.53***	5.51***	2.22**
[5, 9]	−0.89	−0.49	1.22

Note: The table shows the t-value of the Welch's t-test comparing events with positive versus negative sentiment. Events are split into two groups: positive events U with the top 50% by sentiment, and negative events D with the bottom 50%. The sentiment is evaluated on similar sentences, dissimilar sentences, and the normalized difference of them defined in Eq. (5).
* Significance level: (10%).
** Significance level: (5%).
*** Significance level: (1%).

Table 3
Regression coefficients for CAR.

	(1) Large-cap	(2) Small-cap
Constant	9.5*	−11
	(5.6)	(11)
Sentiment (dissimilar)	1.6***	3.3***
	(0.54)	(0.92)
Sentiment (similar)	1.1**	−1.6*
	(0.57)	(0.92)
Mean Similarity	0.025	−0.029
	(3.9)	(6.7)
ΔMargin	12***	9.7***
	(2.5)	(3)
ΔSales	−1.1	4.3***
	(1.2)	(1.2)
LogMKT	−0.11	0.25
	(0.16)	(0.33)
PBR	−0.037	−0.13*
	(0.052)	(0.075)
Observations	1634	1411
Adjusted R^2	0.037	0.027

Note: The table shows coefficients with standard errors in parentheses. The regression formula is described in Eq. (6). The sentiment is evaluated on dissimilar and similar sentences following Eq. (4). The mean similarity is the average similarity across sentences for each event.
* Significance levels: (10%).
** Significance levels: (5%).
*** Significance levels: (1%).

impact on post-event returns. The difference is also confirmed by ΔS in Table 2. In addition, we conduct regression analysis to control for other fundamental factors, periods, and sectors. Table 3 shows the regression coefficients for post-event CARs with standard errors in parentheses. The sentiment of dissimilar sentences has significantly positive coefficients regardless of firm size. By contrast, the sentiment of similar sentences has positive coefficient for large-cap but negative one for small-cap. One interpretation is that sentiment in similar sentences overlaps with fundamental factors, ΔMargin and ΔSales, reducing its explanatory power once controls are included.

Overall, dissimilar sentences have a stronger relation with post-event stock returns than similar sentences, supporting hypothesis. This effect is particularly pronounced for small-cap firms, perhaps reflecting analyst coverage and variety of disclosures other than financial results briefings.

4. Conclusion

Our results show that novel information is processed differently in the market. Dissimilar sentences have stronger explanatory power for post-event CARs than similar sentences. Moreover, we find that statements are incorporated by the market differently depending on their sentiment, novelty and firm size.

Our findings show that textual novelty, beyond sentiment alone, is critical for explaining market responses to corporate disclosures. Future research should extend novelty measurement to news, annual reports, and other documents beyond financial results briefings.

Declaration of Generative AI and AI-assisted technologies in the writing process

During the preparation of this work the authors used LLMs in order to improve the readability and language of the manuscript. After using

this tool/service, the authors reviewed and edited the content as needed and take full responsibility for the content of the published article.

Declaration of competing interest

The authors declare no competing financial or non-financial interests related to the content of this paper

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Data availability

The authors do not have permission to share data.

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